



Software Engineering Department

Braude College of Engineering

Final Project – Phase A (61998)

**Anomaly Detection in Citation Networks via
Deep Structural Learning (GLAD) and
Spectral Analysis**

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Link to GitHub:

<https://github.com/avivraz99/Deep-Graph-Learning-for-Anomalous-Citation-Detection>

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Abstract

Anomaly detection is widely studied in domains such as healthcare, finance, and security, but has received limited attention in scholarly citation networks. Since citations are a central measure of scientific impact and are relatively easy to manipulate, detecting anomalous citation behavior is essential for protecting the integrity of academic evaluation.

This study will investigate structural anomaly detection in citation networks by comparing two complementary approaches. The first will apply a simplified Graph Learning for Anomaly Detection (GLAD) model to identify local structural anomalies by learning typical neighborhood connectivity patterns using Graph Neural Networks (GNNs). The second will employ spectral methods, analyzing the global structure of the network through the principal eigenvector analysis of the adjacency matrix to detect nodes with abnormal spectral influence.

Validation will be conducted on real citation networks augmented with synthetically injected structural anomalies. This controlled setup will enable a systematic comparison between the two approaches by evaluating their ability to detect the injected anomalous nodes. The analysis will focus on detection rates, robustness, and the degree of overlap between the anomalies identified by each method.

1. Introduction

Citations are a cornerstone of academic communication, serving as the main mechanism through which the scholarly influence of papers, researchers, and institutions is evaluated. Maintaining fairness and accuracy in citation-based evaluation is therefore essential to ensure the integrity of scientific research.

Scholarly Network Analysis (SNA) [8] examines the structure of academic networks, and citation network analysis specifically studies how papers are connected through references. With the availability of large-scale academic datasets, citation networks can be modeled as graphs and analyzed using computational methods.

However, common bibliometric indicators, such as the h-index and Journal Impact Factor (JIF), mainly focus on the number of citations, without considering their context or intent. This may encourage behaviors such as excessive self-citation or coordinated citation patterns. These practices can artificially inflate impact measures and create distortions in academic evaluation.

Detecting anomalous citations automatically is a challenging task. Citation networks are complex structures, and many existing approaches focus on identifying unusual papers rather than suspicious citation links. In addition, most methods rely mainly on structural information and do not fully consider the meaning or purpose of citations.

In this Phase A project, the objective is to design a structured and modular framework for detecting citation anomalies. The emphasis at this stage is not on full

implementation, but on clearly defining the model architecture, its components, and the overall research process. This design will serve as a foundation for future development and experimental validation.

2. Background and Preliminaries

To establish a solid foundation for the proposed design, this section introduces the theoretical concepts, formal definitions, and existing approaches relevant to the problem domain.

2.1 Theoretical Foundations

Let $G = (V, E)$ denote a **graph network**, where:

- V is the set of nodes (or vertices), representing entities in the system (e.g., academic papers).
- E is the set of edges (or links), representing relationships between nodes (e.g., citation links).

We define a **citation network** as a directed graph in which nodes represent scholarly articles and directed edges correspond to citations between them [3]. Citation networks are temporal and acyclic: edges flow from citing to cited papers, and the temporal ordering prevents cycles. They exhibit scale-free properties, hierarchical structures, and continuous evolution over time.

The system operates under the following assumptions:

- Assumption 1: Citations can be represented as a network structure with nodes and directed edges.
- Assumption 2: Anomalous citations manifest as deviations from typical structural patterns in the network.
- Assumption 3: Structural embeddings of nodes and edges capture sufficient information to detect anomalies without relying on semantic text.

These assumptions provide the structural framework within which the proposed system is defined

2.2 Existing Approaches

Current methodologies addressing similar problems can be broadly categorized into:

1. **Graph-based approaches:** Methods that learn node or edge embeddings using Graph Neural Networks (GNNs), such as Recurrent GNNs (RecGNNs), Convolutional GNNs (ConvGNNs), Graph Autoencoders (GAEs), and Spatial-Temporal GNNs (STGNNs) [4,5,7].

2. **Data-driven models:** Techniques such as GLAD (Graph Learning for Anomaly Detection) [2,5], which focus on detecting anomalous citations by modeling graph structure and node/edge embeddings.
3. **Spectral methods:** Approaches that leverage eigenvalues and eigenvectors of adjacency or Laplacian matrices to identify anomalies [1].

While these approaches offer valuable contributions, they often suffer from one or more of the following limitations:

- Limited scalability in very large networks.
- Lack of modular structure for integrating multiple detection strategies.
- Insufficient interpretability of learned embeddings.
- High computational complexity, particularly for iterative or deep models.
- Absence of standardized evaluation criteria for anomalous citation detection.

These limitations motivate the need for a structured design that explicitly defines architecture, workflow, and evaluation mechanisms prior to implementation.

2.3 Dual Methodological Framework for the Project

The proposed project adopts a **dual approach** combining:

- **Graph Encoder (GLAD):** A deep graph-based framework that learns latent representations of nodes and edges from the citation network topology. Only the structural component (Citation Graph Encoder) is adopted to detect deviations in connectivity patterns without semantic analysis.
- **Spectral Method with Power Iteration and Normalization:** An iterative algorithm for approximating the dominant eigenvalue and eigenvector of the adjacency matrix, revealing structural anomalies at the global network level.

Both methods are applied in parallel to provide complementary perspectives on anomalous citation detection.

3. Problem Definition and Project Objectives

Problem definition

Although citation networks have been widely studied, there is still no clear and structured framework dedicated to detecting structural anomalies based solely on graph topology.

Most existing approaches either rely on additional semantic information or focus on isolated anomalous nodes without systematically analyzing their structural role within the network. This creates a methodological gap in purely structural anomaly detection.

Accordingly, this project defines the problem as the identification of structurally anomalous nodes in a citation network using only connectivity patterns, without relying on textual or semantic metadata.

Project Objectives

The main objective of this project is to design a dual-method framework for structural anomaly detection in citation networks.

The proposed model combines two complementary analytical perspectives to capture both local and global structural irregularities in the graph.

In addition, the project aims to:

- Examine the differences between the two detection approaches.
- Evaluate their robustness using synthetically injected structural anomalies.
- Analyze the level of agreement between the anomalies detected by each method.

Overall, the goal is to establish a clear and modular framework for structural anomaly detection, which can later be implemented and empirically validated.

4. Proposed Model

The objective of this project is to detect structural anomalies within large-scale academic citation networks. Unlike point anomalies in conventional datasets, a structural anomaly in a graph refers to a node whose connectivity patterns deviate from typical network behavior or from the structure of its local neighborhood.

To address this problem, we propose a dual, comparative framework applying two complementary approaches to the same citation network:

1. **Learning-based method (GLAD-inspired):** Focused on the **Citation Graph Encoder** of the GLAD framework, which uses **Graph Neural Networks (GNNs)** to learn latent structural embeddings of nodes. Only the structural component is implemented, omitting semantic analysis (CPU) to isolate purely topological contributions.
2. **Spectral method (Power Method with normalization):** A classical spectral approach that identifies **global structural anomalies** by analyzing the dominant eigenvector of the adjacency matrix. Nodes with unusually high or low contributions to the principal eigenvector indicate potential global anomalies.

Both methods operate on the same underlying citation graph and produce **node-level anomaly scores**, allowing a direct comparison of **local, learning-based vs. global, spectrum-based anomaly detection**.

4.1 Learning-Based Structural Anomaly Detection: Partial GLAD

A. Graph Construction and Input Representation:

- Citation network is represented as a directed graph $G = (V, E)$, with nodes as papers and edges as citations.
- Each node is initialized with a feature vector based on structural attributes (e.g., citation counts) or uniform initialization.
- The adjacency matrix and node features serve as input to the GNN-based encoder.

B. Citation Graph Encoder:

- Iteratively updates node embeddings through **message passing**, aggregating information from citing and cited neighbors.
- Successive layers encode richer representations of the local neighborhood and the structural role of each paper.

C. Anomaly Detection:

- Node embeddings are analyzed for deviations from typical structural patterns.
- Metrics such as Euclidean distance or cosine similarity are used to assign anomaly scores.
- Nodes with high deviations are flagged as anomalous.

Output:

- Node embeddings matrix and a list of anomalous nodes based on local structural irregularities.

4.2 Spectral Structural Anomaly Detection: Power Method

A. Principal Eigenvector Analysis:

- Operates on the adjacency matrix $A \in \mathbb{R}^{n \times n}$.
- The dominant eigenvector v_1 is computed iteratively using the **Power Method with normalization** [6].
- Nodes with extreme values in

- v_1 indicate **global structural influence**, highlighting potential anomalies

B. Anomaly Scoring:

The mean and variance of the dominant eigenvector are computed to derive the normalized anomaly score (Z-score) for each node i :

$$\mu = \frac{1}{n} \sum_{i=1}^n v_1(i), \sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (v_1(i) - \mu)^2}, AS(i) = \frac{|v_1(i) - \mu|}{\sigma}$$

Nodes with $AS(i) > T_r$ (e.g., $T_r = 2$) are marked as anomalous.

Output:

- Set of nodes exhibiting unusual global structural patterns

4.3 Fusion and Comparative Analysis

- Both methods produce node-level anomaly scores.
- Outputs are aligned using a unified ranking protocol.
- Nodes identified by both methods are considered strong structural anomalies; discrepancies highlight local vs. global deviations.

This dual framework allows a systematic comparison between learning-based embeddings and classical spectral analysis, capturing complementary aspects of structural anomalies.

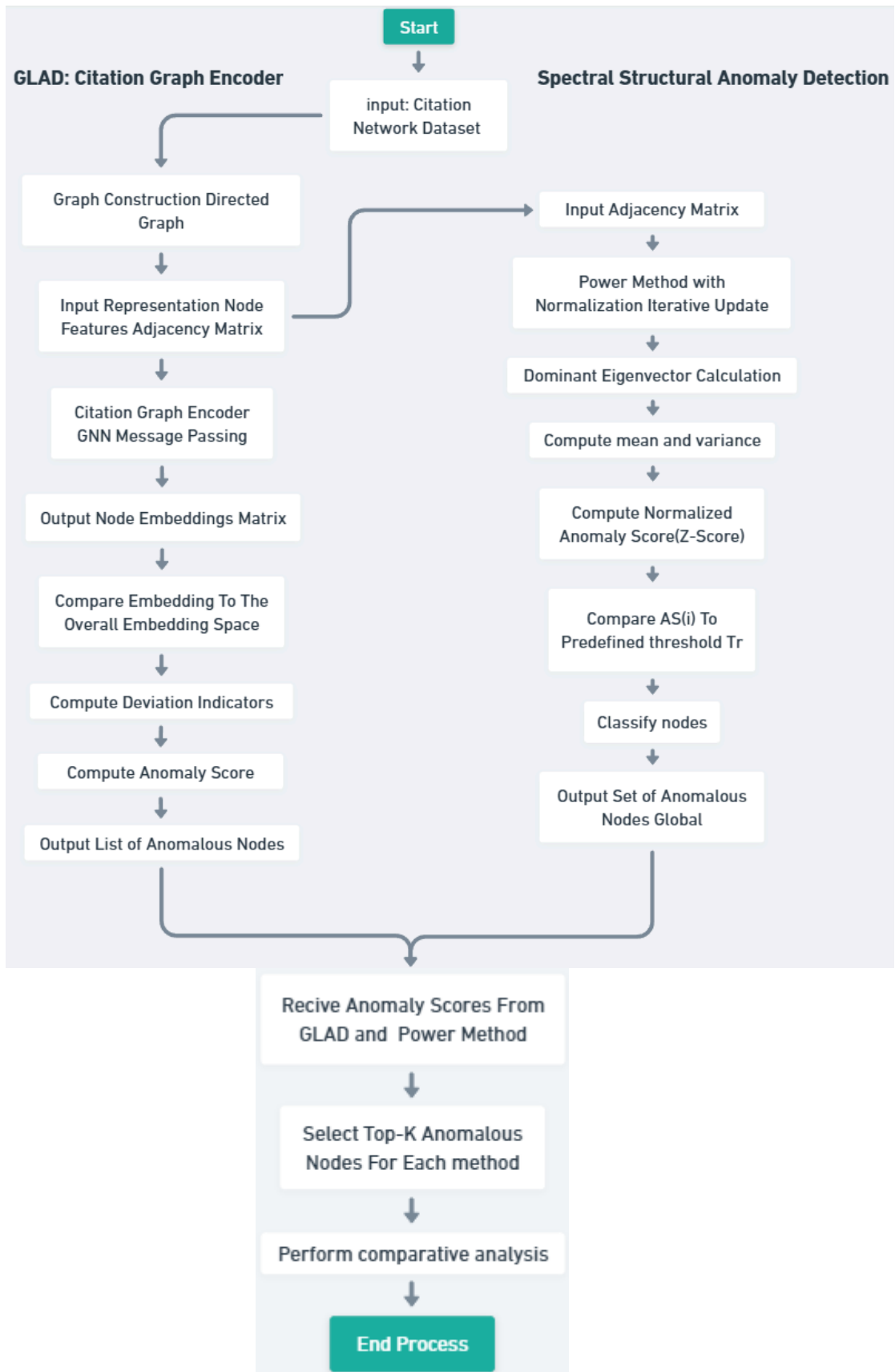


Figure 1. Architecture of the proposed dual-model framework combining a learning-based Citation Graph Encoder and spectral Power Method analysis.

5. Research Process

The research process focused on designing a structured framework for detecting structural anomalies in citation networks.

The work was carried out in four main stages:

1. Literature Review

We reviewed existing work on graph-based anomaly detection, including learning-based and spectral approaches.

This stage helped define what is considered a structural anomaly and motivated the choice of two complementary methods.

2. Method Selection and Definition

Two approaches were selected: a learning-based method and a spectral method.

For each method, we defined the inputs, assumptions, expected outputs, and key structural parameters, without relying on textual data.

3. System Design

A unified dual-approach architecture was designed, where both methods operate on the same citation graph.

The system was structured so that the outputs of both approaches can be compared in a consistent way.

Design considerations included modularity, scalability, and computational feasibility.

4. Validation Planning

A synthetic anomaly injection strategy was defined to allow controlled evaluation in future phases.

Evaluation metrics were planned to compare detection performance and measure agreement between the two methods.

This process establishes a clear design foundation for future implementation and experimental validation.

6. Requirements

The requirements of this project are derived from two complementary sources: the GLAD framework, which provides a structural GNN-based approach for learning typical citation patterns, and spectral analysis methods, which enable the detection of global structural irregularities through eigenvalue analysis. Together, these define a requirements framework suitable for a dual-method design focused on structural anomaly detection.

6.1 Functional Requirements

The system shall:

- Represent each academic paper as a node and each citation as a directed edge in the citation graph.
- Construct the adjacency matrix from the citation network.
- Generate vector embeddings for each node using the GNN-based Citation Graph Encoder (local structural context).
- Update node embeddings iteratively via message passing, aggregating neighbor information.
- Compute node-level structural anomaly scores based on embedding deviations (Euclidean distance, cosine similarity).
- Compute the dominant eigenvector of the adjacency matrix for global spectral anomaly detection.
- Assign Z-score anomaly values to each node and identify nodes exceeding a predefined threshold.
- Support synthetic injection of anomalous nodes or edges for controlled validation.
- Maintain records of injected anomalies for evaluation.
- Produce structured outputs: ranked lists of anomalies for GLAD-based, spectral-based, and combined methods.
- Compute evaluation metrics: Recall@K, Precision@K, and Overlap Ratio@K.

6.2 Not Functional

The system shall:

- Ensure node embeddings accurately represent local connectivity and reflect typical citation behavior (Accuracy, Reliability).
- Scale to large citation networks while maintaining computational stability (Scalability, Performance).
- Produce consistent embeddings across repeated runs (Consistency, Reproducibility).
- Provide interpretable anomaly scores with clear thresholds (Interpretability).

- Allow configuration of thresholds, K-values, and injection parameters without modifying core logic (Maintainability, Modularity).
- Degrade gracefully in edge-case scenarios and maintain robustness under varying synthetic perturbations (Robustness).

The outputs of Phase A include the dual-method system design, functional specifications, node embeddings and adjacency representations, and evaluation metrics definitions, forming the basis for future implementation.

7. Evaluation Framework: Validation and Testing Plan

The dual anomaly-detection system will be evaluated through unit testing, validation, and robustness checks to ensure that each component works correctly and that the overall detection pipeline is stable.

7.1 Validation via Synthetic Anomaly Injection

Since real citation networks lack verified labels for structural anomalies, we will perform controlled validation using synthetic anomaly injection. This involves adding nodes and/or edges with unusual connectivity patterns to the original citation network. The identities of these synthetic anomalies are known and serve as a reference for evaluation.

Both components of the dual model: the learning-based Citation Graph Encoder (GLAD-inspired) and the spectral Power Method, will process the extended network. Each node will receive two scores:

- A local anomaly score from the GNN embedding analysis.
- A global anomaly score from the dominant eigenvector of the adjacency matrix.

Performance will be assessed by checking whether synthetic anomalies appear among the top-K ranked nodes. This setup also allows comparing local vs. global anomaly detection, with nodes detected by both methods considered strong candidates for true structural anomalies.

7.2 Ranking-Based Evaluation Metrics

To quantitatively evaluate anomaly detection performance under synthetic injection, the validation framework suggests employing ranking-based evaluation metrics, specifically Recall and Precision, computed with respect to the top-K highest-ranked nodes according to each anomaly scoring method.

- Recall@K: proportion of injected anomalies among the top-K ranked nodes:

$$Recall@K = \frac{\text{Number of injected anomalies within Top-K}}{\text{Total number injected anomalies}}$$

This metric reflects the model's ability to prioritize true anomalous nodes among the highest-scored candidates.

- Precision@K: proportion of top-K nodes that are true injected anomalies:

$$Precision@K = \frac{\text{Number of injected anomalies within Top-K}}{K}$$

While Recall evaluates detection coverage, Precision assesses the accuracy of the top-ranked anomaly set.

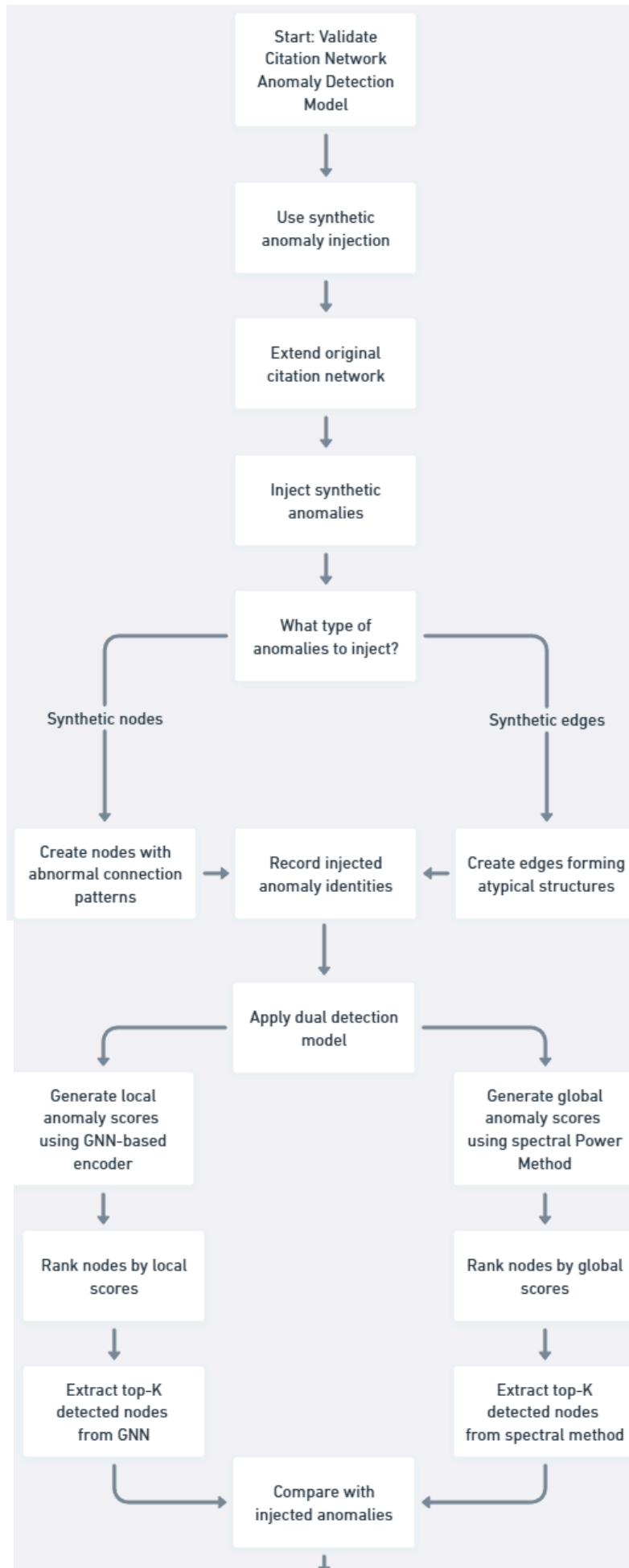
- Overlap Ratio@K: size of the intersection between GLAD-based and spectral top-K anomaly sets divided by their union, measuring agreement between the methods:

$$Overlap Ratio@K = \frac{T_K^G \cap T_K^S}{T_K^G \cup T_K^S}$$

where:

- T_K^G denote the top-K anomaly set produced by the GLAD-based model
- T_K^S denote the top-K anomaly set produced by the spectral method

This enables evaluation of detection coverage, accuracy, and complementarity between the two approaches.



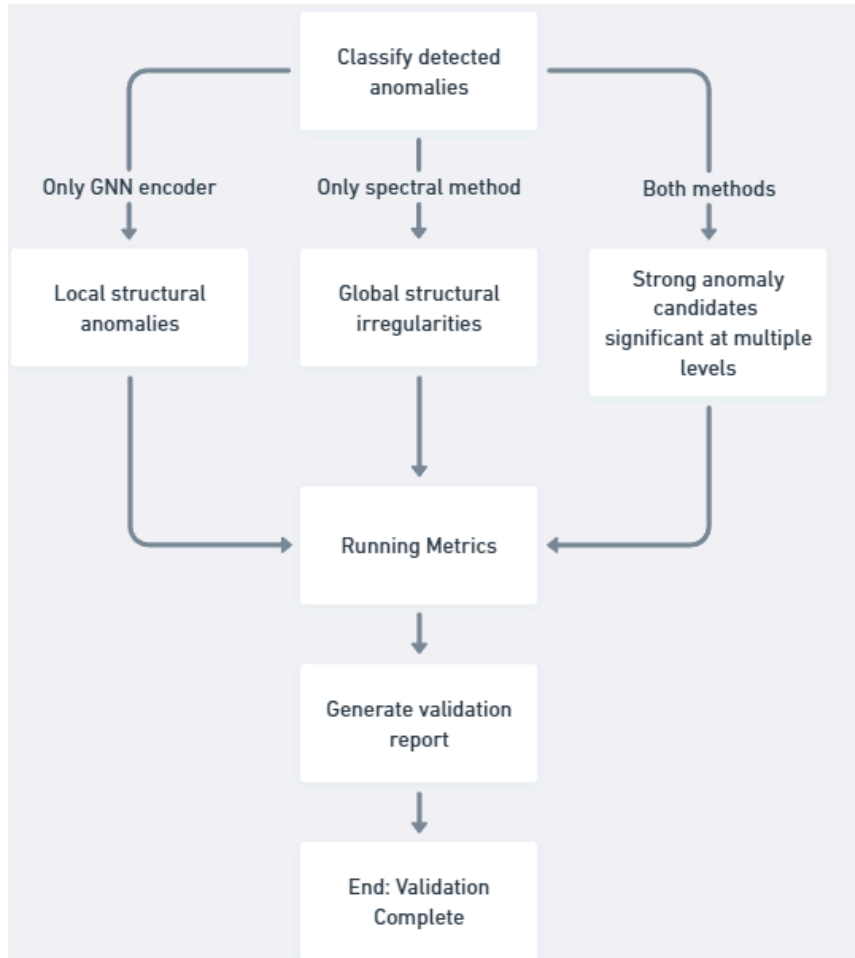


Figure 2. Validation pipeline based on controlled synthetic structural anomaly injection and Top-K ranking evaluation.

7.3 Testing Plan

7.3.1 Unit Testing

- Verify that each component processes inputs correctly, produces valid outputs, and performs calculations as expected.
- Check consistency of scoring, ranking, and aggregation for both pipelines.

7.3.2 Regression Testing

- Maintain a Golden Benchmark dataset with fixed real citation subgraphs and deterministic synthetic anomalies.

- Use it to detect unintended changes in anomaly scores or rankings across development cycles.
- Ensures reproducibility and stability of evaluation results.

8. Anticipated Challenges and Limitations

The design phase of the project involved several conceptual and methodological challenges primarily related to understanding unfamiliar theoretical frameworks and defining a coherent dual-approach solution. These challenges were addressed through an extensive literature review and careful model specification as part of the system design.

The following subsections focus on challenges anticipated in the next phase of the project, namely the implementation and experimental evaluation stage.

8.1 Identifying a Suitable Dataset

A primary challenge in the implementation stage is identifying and selecting an appropriate dataset that satisfies the project's core requirements. First, the project relies solely on structural citation data, while many publicly available datasets emphasize textual features or supervised classification tasks. Second, large-scale graphs introduce computational constraints, particularly for spectral analysis. Finally, data quality issues such as sparsity, missing citations, or incomplete metadata may affect the performance of both learning-based and spectral methods.

8.2 Scalability Considerations

Scalability represents a significant challenge when applying the proposed dual approach to large-scale citation networks with many nodes and edges. Training Graph Neural Networks requires substantial computational resources, while iterative spectral computation over large adjacency matrices may lead to high memory usage and increased runtime.

8.3 Threshold Selection

Determining appropriate anomaly score thresholds presents a methodological challenge, particularly in the absence of ground-truth labels. Fixed thresholds may significantly influence detection outcomes and lead to variability across datasets, making threshold selection a critical factor in the evaluation process.

9. Research Infrastructure and Tools

To implement the proposed anomaly detection system for citation networks, we will utilize tools and technologies that enable graph processing, machine learning, data analysis, as well as document management and knowledge sharing.

Programming Language and Libraries

We will use Python and its scientific libraries to implement the system, including:

- PyTorch (for implementing Graph Neural Networks),
- numpy and SciPy (for numerical computations and matrix operations),
- PyOD (for anomaly detection methods)
- scikit-learn (for general machine learning tools),
- matplotlib (for data visualization).

AI and Language Tools

We use AI tools to assist with text processing, chart making, translation, and writing Gemini, ChatGPT, Whimsical AI, Google Translate, Grammarly.

Cloud Storage and Document Management

To store data, scripts, and reports, we use: Google Drive, Google Docs, and Sheets.

Online Knowledge

To support research and validation, we consult:

Braude Library Catalog (for access to academic papers), Wikipedia (for general knowledge and reference).

Code Repository

All code will be maintained and version-controlled using GitHub (for collaborative development and tracking changes).

10. AI Links

- <https://chatgpt.com/share/697b82ee-d8ec-8010-ac5b-b06ea1008701>
- <https://chatgpt.com/share/697b835c-b5b0-8010-bd52-560bd13d5248>
- <https://chatgpt.com/share/697b840f-662c-8010-8fa4-1356f2267ed5>

- <https://chatgpt.com/share/697ca15a-fd60-8010-9f6e-79b7f67e84c2>
- <https://gemini.google.com/share/1d57f25b5b04>

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Appendices

1. Scholarly Network Analysis

Citation counts are a vital academic measure, yet underlying citation behavior often goes unexamined, sometimes appearing arbitrary or exhibiting a degree of randomness.

Citation-based bibliometric measures are vulnerable to distortions arising from structural and behavioral biases in academic citation practices. For instance, strategic self-citations by authors represent a mechanism for artificially enhancing individual performance metrics (such as the h-index), which can distort the perceived importance of specific research streams and introduce systemic bias favoring existing intellectual paradigms. Self-citations originating from co-authors and collaborative ventures can similarly be misleading and potentially distort the scientific literature.

Reciprocal citation constitutes a pattern wherein authors prefer including work from scholars who have previously cited their own publications, rather than selecting from the broader available literature. This behavior is interpreted not merely as an acknowledgment of prior art, but also as an indication of social or strategic exchange within the scientific community.

Furthermore, journal self-citation practices have been observed, where editors actively encourage or require authors to reference articles previously published within the same journal, ultimately resulting in journal citation stacking. Another related form of citation manipulation is known as coercive or induced self-citation, which can occur among reviewers, grant committees, or academic advisors.

Driven by institutional pressure to sustain or elevate key bibliometric indicators, particularly the JIF and the h-index, some editors engaged in unethical citation practices either by coercing authors or by self-citing their journal content, artificially inflating the respective citation metrics. “Citation cartels” refer to groups of journals exchanging citations excessively. Recent studies have shown that heightened competitive pressures in academic publishing have increased the prevalence of such practices. For example, both Clarivate Analytics and The Scholarly Kitchen frequently report cases of citation cartels. Recipient journals have been documented to boost their JIF by as much as 94% through coordinated citations from donor journals during the relevant Impact Factor period. As bibliometric metrics serve as proxies for journal quality and scholarly standing, accurately identifying and addressing such citation anomalies is essential to maintain indicator validity and ensure proper academic assessment.

Journal citation cartels are more difficult to document and detect than self-citations, primarily due to the challenge of defining "excessive" citation. While network approaches are practical, they can generate false positives because legitimate inter-journal citations within the same field naturally form dense, cartel-like communities.

Although existing literature confirms anomalous citations via network analysis and ethical discussions, automatic detection methods for citation networks are underdeveloped. Some of the current graph anomaly algorithms fail to capture the subtle, pairwise abnormal signals required for this task, and the absence of suitable, labeled datasets inhibits the effective application of supervised machine learning for anomaly detection.

Scholarly networks serve as rich sources of information for analyzing intellectual influence and detecting abnormal interactions such as self-citation fraud or citation cartels.

Methods range from classical metrics like PageRank and betweenness to modern embedding techniques that capture latent features of citation behavior. Temporal and structural modeling of such networks is essential for tracing research trends and identifying outliers.

2. GLAD: Deep Graph Learning for Anomalous Citation Detection

GLAD (Deep Graph Learning for Anomalous Citation Detection) [5] is a deep graph-based learning framework designed to detect anomalous citation behavior in academic citation networks. In this context, a citation network is modeled as a directed graph, where nodes represent scholarly articles and edges correspond to citation links between them.

To capture these aspects, GLAD jointly models three complementary components of the citation network:

- the graph structure, capturing how papers are connected through citations,
- node attributes, representing intrinsic properties of academic papers, and
- edge attributes, describing the characteristics of citation relationships.

GLAD defines a citation as anomalous when it simultaneously exhibits weak semantic relevance between the citing and cited papers and an abnormal relational pattern, such as excessive or socially motivated citations. To support this definition, the framework integrates semantic analysis of citation contexts with structural learning over the citation graph.

GLAD architecture consists of two main components. The first is the Citation Purpose Unit (CPU), which analyzes citation context text in order to infer the intent behind each citation. This module enables the extraction of semantic signals related to citation motivation. The second, and structurally central, component is the Citation Graph Encoder, which learns latent representations of papers and citations directly from the network topology and associated features.

The Citation Graph Encoder employs graph neural networks (GNNs) to perform message passing over the citation graph. Through this process, each node iteratively aggregates information from its neighbors, allowing the model to learn embeddings that reflect typical citation connectivity patterns and the structural role of each paper within the network.

Node and Edge Representation Learning (conceptual)

The Citation Graph Encoder produces latent vector representations for both nodes (papers) and edges (citations). Node embeddings capture the local neighborhood structure and each paper’s structural role within the citation graph, reflecting typical connectivity patterns. Edge embeddings summarize relational properties between citing and cited papers, such as common co-citation patterns or structural importance. These learned representations form the basis for identifying structural anomalies: nodes or edges whose embeddings deviate significantly from normative patterns are considered potentially anomalous.

While GLAD was originally designed as a comprehensive framework combining semantic and structural analysis for edge-level anomaly detection, its graph-based encoding mechanism also provides a strong foundation for studying structural anomalies in citation networks. This capability motivates its partial adoption in the proposed project, where only the structural component, the Citation Graph Encoder, is implemented to focus on learning connectivity patterns for anomaly detection.

This conceptual overview of GLAD highlights the central role of the Citation Graph Encoder in learning structural representations from citation networks. While the full GLAD framework integrates semantic context for edge-level anomaly detection, in the present project, we adopt only the structural encoding component. This allows us to focus on the core task of capturing typical citation connectivity patterns and detecting deviations, which is operationalized in the project’s implementation described in Chapter 4.

3. Power Method with Normalization

The Power Method with normalization [6] is an iterative algorithm used to approximate the dominant eigenvalue (i.e., the eigenvalue with the largest absolute value) of a matrix and its corresponding eigenvector. The method repeatedly multiplies a vector by the matrix and normalizes (scales) the result at each iteration.

The normalization step prevents numerical overflow or underflow and supports convergence toward the dominant eigenvector.

Power Method with Normalization Algorithm

1. Choose an initial non-zero vector $x^{(0)}$ and normalize it $\overline{x^{(0)}} = \frac{x^{(0)}}{\|x^{(0)}\|_\infty}$
2. Define an iterative sequence for $m=1,2,3,\dots$;

$$x^{(m)} = A * \underline{x}^{(m-1)}$$

2.1 At each iteration, compute $x^{(m)}$ and its norm $S_m = \|x^{(m)}\|_\infty$

2.2 Compute the normalized vector $\overline{x^{(m)}} = \frac{x^{(m)}}{S_m}$

Under standard assumptions (in particular, that the dominant eigenvalue is unique and strictly greater in magnitude than the others), the sequence converges to the dominant eigenvector $v^{(1)}$ and the scaling factors S_m converge to the dominant eigenvalue:

$$\lim_{m \rightarrow \infty} \underline{x}^{(m)} = v^{(1)} \qquad \lim_{m \rightarrow \infty} S_m = \lambda_1$$

Due to its simplicity and scalability, the Power Method with normalization is particularly suitable for analyzing large adjacency matrices arising from citation networks.